

Inverse problems in biological imaging

Homework Assignment 2

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Problem 1

We consider the following optimization problem:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|_2^2 + \lambda \mathcal{R}_{\alpha, \beta}(x) + i_{\geq 0}(x), \quad (1)$$

where

$$\mathcal{R}_{\alpha, \beta}(x) := \|x\|_1 + \langle \alpha, x \rangle + \beta, \quad \alpha \in \mathbb{R}^n, \beta > 0 \quad (2)$$

and A is a convolution operator: $Ax = h \star x$.

Part 1

First, we are going to derive the proximal operator of $\mathcal{R}_{\alpha, \beta}$:

$$\text{prox}_{\gamma \mathcal{R}_{\alpha, \beta}}(y) = \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y\|_2^2 + \gamma \mathcal{R}_{\alpha, \beta}(x) \right\} \quad (3)$$

$$= \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y\|_2^2 + \gamma \|x\|_1 + \gamma \langle \alpha, x \rangle + \gamma \beta \right\} \quad (4)$$

$$= \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y\|_2^2 + \gamma \|x\|_1 + \gamma \langle \alpha, x \rangle \right\} \quad (5)$$

$$= \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y\|_2^2 + \gamma \|x\|_1 + \langle \gamma \alpha, x \rangle + \langle \gamma \alpha, -y \rangle + \frac{1}{2} \|\gamma \alpha\|_2^2 \right\} \quad (6)$$

$$= \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y\|_2^2 + \gamma \|x\|_1 + \langle \gamma \alpha, x - y \rangle + \frac{1}{2} \|\gamma \alpha\|_2^2 \right\} \quad (7)$$

$$= \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - y + \gamma \alpha\|_2^2 + \gamma \|x\|_1 \right\} \quad (8)$$

$$= \arg \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - (y - \gamma \alpha)\|_2^2 + \gamma \|x\|_1 \right\} \quad (9)$$

$$= \text{prox}_{\gamma \|\cdot\|_1}(y - \gamma \alpha) \quad (10)$$

$$= \text{ST}_{\gamma}(y - \gamma \alpha). \quad (11)$$

We showed that the proximal operator of $\mathcal{R}_{\alpha, \beta}$ is a shifted soft threshold function.

Part 2

The following pseudo-code describes iteration $k + 1$ of the FISTA algorithm for the optimization problem above. With $\widehat{x}$ we denote the Fourier transform of x and with $\mathcal{F}^{-1}(z)$ the inverse Fourier transform of z .

Algorithm 1 FISTA Iteration

Require: x_k, x_{k-1} two previous estimates, t_k previous momentum scalar, τ step size

- 1: $t_{k+1} \leftarrow \frac{1 + \sqrt{1 + 4t_k^2}}{2}$
 - 2: $y_k \leftarrow x_k + \frac{t_k - 1}{t_{k+1}}(x_k - x_{k-1})$
 - 3: $x_{k+\frac{1}{2}} \leftarrow y_k - \tau \mathcal{F}^{-1}(\widehat{h}^* \odot (\widehat{h} \odot \widehat{y}_k - \widehat{y}))$
 - 4: $x_{k+1} \leftarrow \max(0, \text{ST}_{\tau\lambda}(x_{k+\frac{1}{2}} - \tau\lambda\alpha))$
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Part 3

The parameter β is irrelevant for the described optimization problem since the contribution of β to the objective function is constant in x .